

3D single pixel imaging based on parallel measurement with quadrant detector

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ABSTRACT

Structured light three-dimensional (3D) imaging has advantages such as high accuracy, high resolution, and non-contact, and has enormous application value in fields such as automotive manufacturing and cultural relic detection. However, it often requires multiple structured light encoding to obtain 3D information, thus limiting the speed of 3D imaging. Single pixel imaging (SPI) technology, due to its use of structured light and single point detection to jointly obtain image information, can simply achieve simultaneous detection of multi-dimensional information through a single pixel detector array. Therefore, the structured light 3D imaging technology is combined with the single-pixel technology of multi-channel quadrant sensing, and the modulation of three structured light fields of red, green, and blue light is achieved separately through the decoupling of spatial 3D information and spectral dimension information. Combined with a quadrant sensing detector integrated with red, green, and blue filtering, simultaneous measurement of three structured light field signals is achieved. Thus, a scheme demonstration is accomplished to improve the imaging speed of 3D imaging by three times through decoupling. Further combining Gray codes and optimizing Hadamard sequences using compressive sensing ensures the accuracy and imaging quality under undersampling of 3D imaging. The experimental results show that the RMSE of our method is only 0.0576 mm. This method can be further extended to achieve high-precision and high-quality 3D reconstruction using more channel structured light modulation and more spectral detector arrays in only one parallel measurement.

1. Introduction

Three dimensional (3D) imaging, as a means of information acquisition to perceive the real 3D world, has attracted people's interest and attention in a wide range of fields, such as cultural relic protection, scientific research, security, automotive autonomous driving, biomedical engineering, and entertainment [1–5]. Different principles and methods are utilized for 3D imaging, including time-of-flight [6], photogrammetry measurement [7], and photometric stereo [8]. The high-speed 3D measurement technology can obtain the 3D topography information

of dynamic objects under the condition of high-speed motion, and can analyze high-speed scenes such as broken objects, vibrating objects or rotating objects [9,10], which has very important significance in industrial fields and scientific research. Because of the advantage of sinusoidal phase-shift method in precision, phase-shift method has been widely used in optical 3D measurement [11,12]. To realize a 3D measurement by using phase shift method, it is generally necessary to project multiple patterns to the scene without dynamic changes, which makes it have no speed advantage compared with the 3D measurement method of single frame projection. Therefore, some new technical means are needed

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to reduce the number of structured light fields and strive to achieve the performance of 3D imaging with single modulation detection.

In order to improve the speed (reduce the amount of structured light modulation required) of 3D measurement based on sinusoidal phase shift method, many researches have been done. Specifically, using the three-frequency heterodyne method (multi-wavelength method), Wang and Zhang [13] projected binary fringes at a projection rate of 5000 Hz to achieve a reconstruction speed of 556 Hz. They then used the dual-frequency method to measure the 3D surface morphology of the beating isolated rabbit heart with a total of 6 patterns of “3+3”, which improved the measurement efficiency [14]. Zuo et al. [15] put forward the Bi-frequency method based on the number theory method, reducing the projection pattern to 5 (number theory, “3+2”). How to simultaneously modulate and receive multiple structured lights and signals required for 3D imaging to achieve one single projection and detection of 3D imaging (similar to single frame 3D imaging in mode) is crucial for improving the speed of high-precision 3D imaging.

In ensuring the accuracy of 3D imaging measurement, researcher introduce Gray code assisted phase shift technology to carry out phase unwrapping. The unique coding characteristics of Gray code patterns can be used to eliminate ambiguity of truncated phase. In order not to lose the signal-to-noise ratio of Gray code, Zhang et al. [16] conducted research on improving the coding utilization rate, and proposed cyclic complementary Gray code, using a total of 7 patterns of “3+4” to achieve a 3D shape reconstruction frame rate of 310 frame/s. They later presented the Gray code multiplexing encoding strategy, using “3+1” four patterns to achieve a 3D shape reconstruction frame rate of 452 frame/s. Furthermore, they [17] proposed a structured light multiplexing encoding strategy, successfully achieving a reconstruction rate of 3174 frames per second for “2+1” patterns. Hence, the introduction of Gray code effectively guarantees the accuracy of 3D imaging. If multiple structured lights and signals are modulated and received simultaneously in one single projection detection, the introduction of Gray codes in 3D imaging will provide practical and effective guarantees for achieving high-precision 3D imaging with single modulation and reception.

Noteworthy, the development and progress of SPI technology, which can obtain image information by using only one single pixel detector and spatial light modulation equipment, has made it possible to solve the above problems. Due to its advantages of wide spectrum and low cost in special bands, it is expected to make up for or replace the existing imaging technology in special scenes [18–21], including 3D imaging. Recently, the research work on single-pixel 3D imaging has also received extensive attention. Such as, the 3D SPI of multiple single-pixel detectors (multi-view) using photometric stereoscopic technology [22–24]; time-of-flight [25–28]; Fourier transform profilometry [29]. These work further proves the effectiveness and superiority of SPI in 3D information acquisition. It is worth noting that the research work of single-pixel 3D imaging based on four-step phase-shifted structured light to obtain 3D information has also been reported Ma et al. [30], which lays a foundation for SPI technology to solve the above single detection 3D imaging problems. It has been demonstrated that utilizing multiple integrated optically filtered single-pixel detectors or array detectors along a single optical path, in conjunction with multiple spatial light modulation devices, can significantly enhance optical modulation and detection efficiency [31] as well as improve the speed of 3D imaging. Notably, for 3D imaging techniques that employ only three sets of structured light fields, this method requires just one experiment to complete 3D imaging, thereby greatly increasing imaging speed. This advancement can further promote the application of SPI technology in 3D imaging-related fields and is anticipated to become widely adopted in the future for dynamic 3D shape measurement of complex scenes, including remote sensing, size measurement and shape analysis of components in industrial assembly lines, and collision testing in the automotive industry.

Fortunately, one single optical path single pixel multi-spectral imaging scheme using only one detector array is proposed [31], which provides further possibilities for the development of this study. Based on

previous work, by combining spatial dimension with spectral dimension decoupling, Gray code and phase shift techniques, we presented and demonstrated a single pixel 3D imaging scheme for parallel measurement of one single optical path. Combined with the use of compressed sensing (CS) [32,33] and cake cutting Hadamard optimization sequence [34–36], the 3D image quality under under-sampling rate is effectively guaranteed. These measures enhance the performance of 3D imaging, enabling accurate 3D reconstruction with a single modulation and the detection of phase shift maps in suitable experimental setups. However, for colored 3D objects, the RGB light projected onto their surfaces cannot be accurately distinguished by the three-channel single-pixel quadrant detector based on wavelength. Additionally, incident light is predominantly absorbed by black or dark-colored objects, which leads to a reduction in the intensity of reflected light and results in poor imaging quality. Furthermore, due to the working mechanism of the digital micromirror device (DMD), only binary patterns can remain stable at any given time. Therefore, to simultaneously collect data from all three channels, a three-channel optical filtering and quadrant sensing single-pixel sensor must be equipped with a DLP structured light projector that contains three DMDs internally. However, the DLP4500 used in this study is equipped with only one DMD, making this experiment purely demonstrative. In future experiments, utilizing a DLP structured light projector with three DMDs will facilitate simultaneous data collection from all three channels. This limitation should be acknowledged as part of the current experiment.

2. System and principle

The basic framework of the 3D imaging method proposed in this paper consists of two parts: The first is a digital light projection (DLP) system which can simultaneously modulate multiple structured light fields and distinguish structured light by different wavelengths; The second is a multi-spectral SPI system with the decoupling of 3D information and spectral information to realize simultaneous detection of multiple structured light fields modulated by the DLP system.

In the first part, the main function of the DLP system is to achieve simultaneous modulation and projection of multiple structured light fields for 3D imaging of structured light onto the 3D target object, in order to achieve a single projection of all required 3D imaging structured light fields. For colored 3D objects, the RGB light projected onto their surfaces cannot be accurately distinguished by the wavelengths using the three-channel single-pixel quadrant detector. Additionally, most black or dark-colored objects absorb incident light, resulting in lower reflected light intensity, which leads to poor imaging quality. The white 3D target can be distinguished by wavelength, for instance, red, green, and blue are illuminated onto their respective modulator components by the DLP system of three liquid crystal displays (3-LCD) or 3-DMD, and then the light is combined. This can achieve the synthesis of three separately modulated structured light fields into a beam and then irradiate it onto a three-dimensional target. Due to being three different optical bands, the modulated light fields do not interfere with each other.

The phase unwrapping method (Gray code-assisted phase-shift techniques) is used here to obtain 3D information, with the phase unwrapping pattern projected onto the target object by the DLP. Because of its advantages in accuracy, the sinusoidal phase-shift method has been widely used in optical 3D measurements [11,12]. The phase-shifting technique [37] is a method that utilizes a digital projector to uniformly shift sinusoidal fringes N times within one cycle ($N \geq 3$), with each shift being $\frac{2\pi}{N}$ radians. In order to obtain high precision 3D topographic information of the target object, a 12-step phase shift is used in our experiment. The sine function expression for the 12-step phase shift stripe is as follows:

$$P_n(x, y) = A(x, y) + B(x, y) \cos \left[\phi(x, y) + \frac{2\pi(n-1)}{12} \right], \quad (1)$$

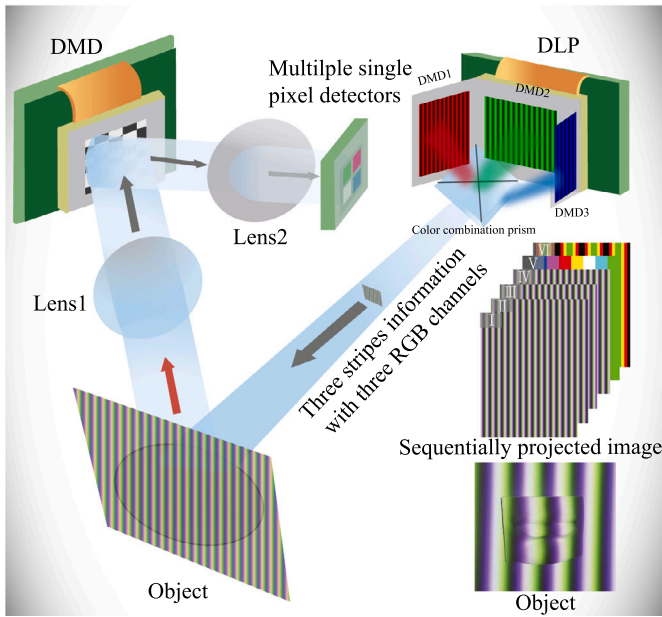


Fig. 1. Schematic diagram of the proposed system. Inside the DLP consists of three DMDs, each of which projects a grayscale map, three stripes at a time, which are converged into a color stripe map by a color combination prism as a projection branch onto the object surface. As imaging branches, Lens1, DMD, Lens2 and Multichannel detector constitute the subsystem of passive SPI.

where, $x = 1, 2, \dots, r$, $y = 1, 2, \dots, c$, $n = 1, 2, \dots, 12$, $A(x, y)$ represents the background light intensity, and $B(x, y)/A(x, y)$ represents the contrast of the fringes. The phase shift fringe generated by Eq. (1) is depicted in Fig. 2(a). Making use of the following equation:

$$\phi(x, y) = \arctan \left[\frac{\sum_{n=1}^{12} P_n(x, y) \sin(2n\pi/12)}{\sum_{n=1}^{12} P_n(x, y) \cos(2n\pi/12)} \right], \quad (2)$$

the wrapped phase $\phi(x, y)$ carrying the surface shape information of the object is computed for these 12 deformed frames of the fringes. In Fig. 2(e), the blue line ϕ wrapped is the value of the wrapped phase solved by Eq. (2), and it can be seen that the wrapped phase is truncated in each period, so the Gray code is utilized to assist in solving the phase, enabling the acquisition of the absolute phase [38].

In order to achieve robust measurements in high-noise environments, researchers have introduced Gray codes to assist in phase unwrapping through phase-shifting techniques [38]. Gray code, initially presented by Frank Gray at Bell Labs in the 1940s [39], dictates that within a set of encoded numbers, any two adjacent code words should differ by only one binary digit (Hamming distance of 1). Furthermore, due to the fact that the maximum and minimum values of its encoding differ by only one digit, creating a “cyclic” pattern, it exhibits strong robustness and noise resistance.

Due to the non-uniformity in the surface reflectivity of the measured object and the intensity of the background light, compounded by factors such as noise, the sharpness of the grayscale code black-white boundaries is compromised, making it challenging to achieve an ideal binary distribution. Consequently, after binarization of the images, mismatches may occur between the decoded level edges and the truncated phase edges, thereby leading to errors in phase unwrapping. To address this issue, the complementary Gray code method was employed [40]. Fig. 2(b) and 2(c) show four traditional Gray codes and one complementary Gray code respectively, making a total of five Gray codes.

Using Eq. (1), the fringes generated in our experiment consist of 16 cycles of sinusoidal patterns, as illustrated in Fig. 2(a). To unwrap truncated phase into absolute phase, 16 phase orders are needed for unwrapping. The four Gray code patterns are precisely used to mark 16

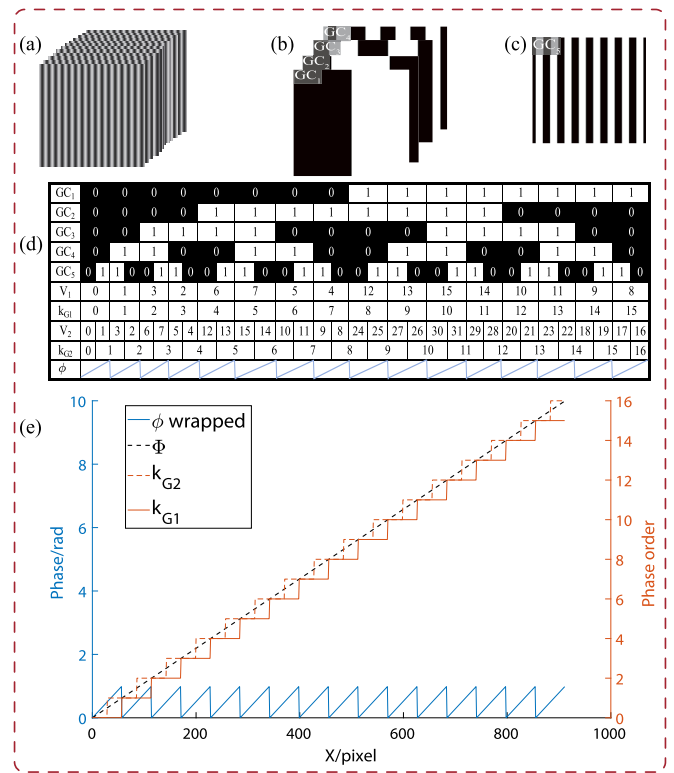


Fig. 2. Gray code-assisted phase shifting technique. (a) Projected phase-shifted fringes; (b) Projected Gray code fringes; (c) Projected complementary Gray code fringes; (d) Solving the Phase order process of complementary Gray codes. (e) Schematic of the absolute phase Φ , which is obtained by solving the phase level k_{G1} solved by four conventional Gray codes and the phase level k_{G2} solved by four conventional Gray codes plus one complementary Gray code expanding the wrapped phase ϕ .

phase orders, corresponding to $2^4 = 16$ phase orders. First, the phase orders are calculated using the first four traditional Gray codes as k_{G1} :

$$V_1(x, y) = \sum_{i=1}^4 GC_i(x, y) \times 2^{(4-i)}, \quad (3)$$

$$k_{G1}(x, y) = i(V_1(x, y)), \quad (4)$$

where V_1 denotes the phase order of the traditional Gray code solution, and GC_i represents the codeword corresponding to the i^{th} Gray code image. k_{G1} is the mapping table, which maps the orders in V_1 to the incremental sequences.

Then, four traditional Gray codes plus one complementary Gray code are utilized, resulting in a total of five Gray codes to obtain k_{G2} :

$$V_2(x, y) = \sum_{i=1}^5 GC_i(x, y) \times 2^{(4+1-i)}, \quad (5)$$

$$k_G(x, y) = i(V_2(x, y)), \quad (6)$$

where V_2 is the phase sequence solved by adding complementary Gray codes, and GC_i represents the codeword corresponding to the i^{th} Gray code image. k_{G2} is the increasing order mapping table of V_2 .

As illustrated in Fig. 2(d), the level edges of k_{G1} and k_{G2} exhibit precise staggering. Therefore, when unfolding the truncated phase, Eq. (7) only uses the code words of the middle part of k_{G1} and k_{G2} to calculate the different parts of the wrapped phase ϕ respectively to obtain the absolute phase Φ .

$$\Phi(x, y) = \begin{cases} \phi(x, y) + 2\pi k_{G2}(x, y), & \phi(x, y) \leq -\frac{\pi}{2} \\ \phi(x, y) + 2\pi k_{G1}(x, y), & -\frac{\pi}{2} < \phi(x, y) < \frac{\pi}{2} \\ \phi(x, y) + 2\pi k_{G2}(x, y) - 2\pi, & \phi(x, y) \geq \frac{\pi}{2} \end{cases} \quad (7)$$

This approach prevents the generation of edge jump errors from the source. Consequently, errors can be avoided when the width of the error-prone region at the edges is less than half the width of a stripe cycle. Additionally, since this method preemptively avoids error generation, it remains unaffected when the object itself contains steep transition regions. As depicted in Fig. 2(e), the blue line represents the wrapped phase ϕ solved using phase shift stripes. Noticeably, the wrapped phase exhibits truncation. The orange solid line and the orange dashed line depict the phase levels k_{G1} and k_{G2} resolved using conventional Gray codes and complementary Gray codes, respectively. It is evident that k_{G1} and k_{G2} are precisely staggered by half a cycle. This staggering aids in resolving potential jump errors occurring at the edges when determining the absolute phase. Additionally, the black dashed line represents the absolute phase Φ of the final solution. It is notable that a continuous, monotonically increasing absolute phase is achieved.

In the second part, i.e., the proposed the multi-channel parallel acquisition imaging system. The structural diagram of the system is depicted in Fig. 1, which is divided into two branches: one is the lighting branch, the other is the imaging branch. In the lighting branch, the DLP composed of multiple independent optical band modulation DMDs is used as an active light field modulation device to project the structured light stripe patterns required for 3D imaging onto the surface of the object. For example, in Fig. 1, the three DMDs are modulated by the red, green, and blue optical bands, allowing for the simultaneous modulation of three independent structured light fields. In the multi-spectral imaging branch, the image of a three-dimensional object passing through the acquisition lens is modulated by a preloaded spatial light field for DMD encoding. The encoded information is collected through a four quadrant detector integrated with red, green, and blue filters. Among them, the four quadrant detector is placed in the optical area where the object and all encoded information pass through after collecting the lens, ensuring that the detection images in each band of the four quadrants maintain good uniformity and can better record phase deformation information.

In this part, we introduce the novel application of single-pixel detector technology with three-channel quadrant sensing to the RGB three-channel fringe projection acquisition process, achieving a threefold increase in imaging speed. To tackle the substantial computational demands associated with SPI process, we employ a CS theory to reduce the sampling rate. The cake-cutting Hadamard basis sort illumination patterns $I_m(x, y)$ are coded by a DMD. The phase encoding projected by DLP onto a three-dimensional object $T(x, y)$ can be represented as:

$$P_k^\lambda(x, y), \lambda = R, G, B, \quad (8)$$

where, $k = 1, 2, \dots, K$, here K is the upward rounding of $\frac{N}{3}$, where 3 is the number of parallel projections of the three structured RGB light fields simultaneously projected. Therefore, the projected structured light field of DLP involves 12 phase-shift coding, four Gray codes and one complementary patterns, so $K = 6$. And according to Eq. (1), the phase representation can be mapped to:

$$P_1^R(x, y) = P_1(x, y), \quad (9)$$

$$P_1^G(x, y) = P_2(x, y), \quad (10)$$

$$P_1^B(x, y) = P_3(x, y). \quad (11)$$

Hence, the probe value can be expressed as:

$$y_m^\lambda = \iint I_m(x, y) \cdot P_k^\lambda(x, y) \cdot T(x, y) dx dy, \quad (12)$$

$$y_n = \iint I_m(x, y) \cdot P_n(x, y) \cdot T(x, y) dx dy. \quad (13)$$

And, the cake-cutting Hadamard basis sort illumination patterns vectors $\{\Phi\}_{m=1}^M$, each row of the matrix Φ can be transformed into a square two-dimensional light field matrix, i.e. the first row of ϕ is converted to the two-dimensional matrix as $I(x, y)$. If a two-dimensional object $T(x, y)$ is converted to a one-dimensional column vector $\mathbf{x}$, the relationship between the probe value $\mathbf{y}$, the light field Φ , and the object $\mathbf{x}$ can be expressed as:

$$\mathbf{y} = \Phi \mathbf{x}, \quad (14)$$

where Φ is a $M \times Z$ ($Z = r \times c$) measurement matrix. However, the Eq. (14) cannot be gained by classical matrix inversion because the number of equations is much smaller than the number of unknown variables. For addressing the issue, the principle of CS theory is relied upon. CS enables the SPI system to decrease the number of measurements required for reconstruction thereby reduction in data storage and data transfer requirements. It can significantly enhance optical modulation and detection efficiency, as well as improve the speed of 3D imaging. Particularly for the 3D imaging technique that utilizes only three sets of structured light fields, this method requires just one experiment to complete the 3D imaging, which greatly increasing imaging speed. This work advances SPI technology in 3D imaging and could be applied to dynamic 3D shape measurement in fields such as remote sensing, industrial component analysis, and vehicle collision testing. Consequently, the signal $\mathbf{x}$ can be perfectly restored by TVAL3 method with linear programming from $\mathbf{y}$, even if $M \leq Z$. TVAL3 is proposed by [41]. This model of the method is based on total variation (TV) minimization, as shown in the following equation:

$$\min_s \sum_i^n \|D_i \mathbf{x}\|, \text{ s.t. } \Phi \mathbf{x} = \mathbf{y}, \quad (15)$$

where $\|\cdot\|$ can be either ℓ_1 or ℓ_2 . $D_i \mathbf{x}$ is the discrete gradient of $\mathbf{x}$ at pixel i . Since this problem can be easily transformed into a quadratic program with convex constraints, there are many easy-to-handle and efficient optimization methods. Thus, the image information after the interaction of the previous phase shift, Gray code, and complementary image with the three-dimensional object can be obtained through Eqs. (11)-(15). Then use Eqs. (2)-(7) to calculate the obtained three-dimensional image information of the object and achieve three-dimensional imaging. To obtain the 3D surface information of the object, the absolute phase must be converted to the height using the phase-to-height algorithm. In the phase-height models, a reference plane with height $h(x, y) = 0$ is required. The object's height is then measured relative to the position of the reference plane. The phase plays a role that relates to the height directly. Specifically, the height can be a function of a phase difference, which is the difference between the object's phase and the reference plane's phase, i.e., $h(x, y) = f(x, y, \Delta\phi, \theta)$, where θ is a vector that consists of parameters to be calibrated. Alternatively, the height can also be the function of the object's phase itself, i.e., $h(x, y) = f(x, y, \phi, \theta)$.

In the phase-height mapping model, the polynomial phase-height calibration model demonstrates the nonlinear relationship between height $h(x, y)$ and phase difference $\Delta\phi(x, y)$. This model does not impose strict geometric constraints on the system configuration, as illustrated by the following formula:

$$h(x, y) = \sum_{i=0}^n a_i(x, y) \Delta\phi(x, y)^i, \quad (16)$$

where $a_i(x, y)$ is the pixel-wise coefficient to be calibrated. The coefficients are often saved after the calibration and then loaded during the 3D measurement. Actually, the effect of lens distortion on the surface reconstruction can be relieved by increasing the degree of polynomial. However, n should be carefully adjusted as Runge phenomenon may occur when a high-degree polynomial function is used. In this case, system calibration was performed using a 3rd order polynomial, with the calibration board calibrated at 11 positions respectively, each time by

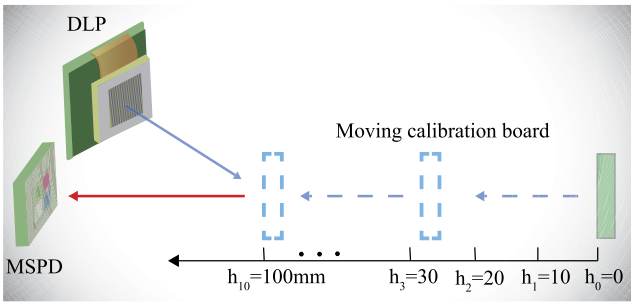


Fig. 3. The process to calibrate the measurement volume for the phase-height models.

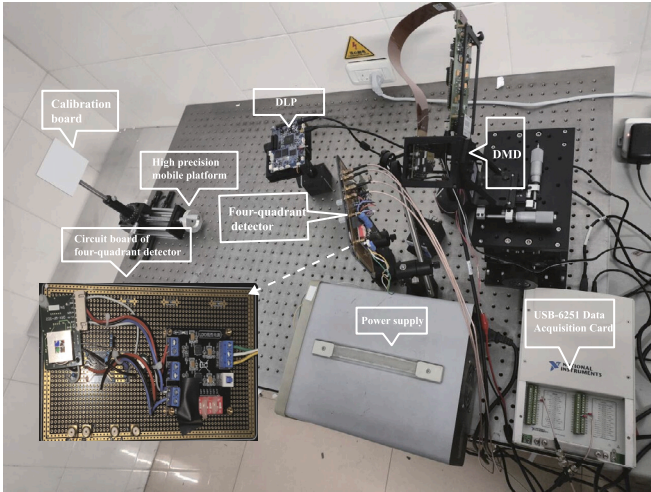


Fig. 4. The experimental setup.

10 mm. The schematic diagram of the system calibration is displayed in Fig. 3.

3. Experiments and results

Experiments have been conducted to test the performance of our proposed method. The experiment setup is shown in Fig. 4. The DLP (LightCrafter 4500, 912×1140 pixels, 4,225 Hz (1 bit)) is placed at a distance of 475 mm from the object and projects the phase-shifted structured light field onto the target object. The reflected signal displays its image information through the lens on the optical modulation plane of the DMD (V-7001, 1024×768 pixels, 22,272 Hz (1 bit)). The imaging lens has focal length of 30 mm and the working distance of the DMD is 875 mm. Then, the signal light after DMD modulation is converged by the collecting lens and illuminated to the four-quadrant detector (First Sensor, QPS0-6, photodiode with $4 \times 12 \text{ mm}^2$ active area, 50 mm^2 Quadrant PIN detector) with integrated the RGB optical filtering. And the USB-6251 Data Acquisition Card simultaneously collects the four-quadrant detector intensity signals from the RGB three channels and transmits them to the local computer for further processing.

In all the experiments, 12 sinusoidal stripes and four traditional Gray codes plus a complementary Gray code were projected. The resolution of our experimental imaging is 256×256 pixels. Full sampling would require 65,536 times, but with the utilization of the Cake-cutting Hadamard basis sort illumination patterns combined with the TVAL3 algorithm, higher image quality was achieved with only 10,000 samples, resulting in a sampling rate of 15%. In the simultaneous reconstruction of three stripe images using the Multiple single pixel detectors, the DMD will refresh only 10,000 patterns. At the same time, the Multiple single pixel detectors will synchronously collect $10,000 \times 3$ light inten-

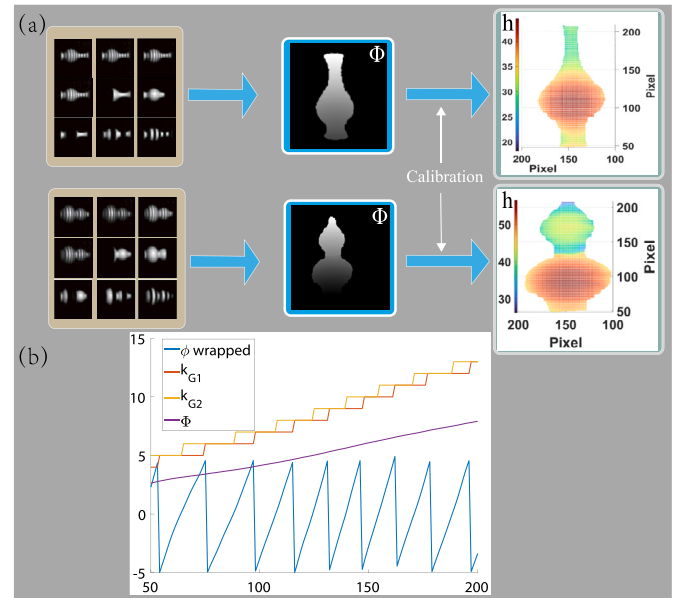


Fig. 5. The process of 3D reconstruction. (a) On the far right are the partial phase-shifting stripes of a white vase (top) and a white gourd (bottom) reconstructed and four traditional Gray codes plus one complementary Gray code, in the middle are the unwrapped absolute phases, and on the far right is the 3D reconstruction result. (b) The phase unwrapping process of a certain row of data from the first model in Fig. (a).

sity signals. Absolute phase recovery requires twelve sinusoidal stripe maps plus four plus Gray code maps and one complementary Gray code map, so the DLP needs to project six times ($\text{int}[(12 + 4 + 1)/3] = 6$). Therefore, to reconstruct the 3D shape of an object, DMD only needs to refresh the Cake-cutting Hadamard basis sort illumination patterns $6 \times 10000 = 60000$ times. The three-channel single-pixel detector reduces the number of projected fringe patterns by a factor of three, while the TVAL3 algorithm decreases the sampling rate for image reconstruction to approximately 15%. This significantly enhances the speed of 3D reconstruction, making it a promising solution for complex, high-speed measurement scenarios, such as measuring the shape and size of components on industrial production lines and conducting automotive crash tests.

3D Reconstruction of Simple Objects. The first experiment included the reconstruction of a white marble vase and a white gourd to observe the effect of 3D reconstruction and verify the proposed method. As shown in Fig. 5(a), on the far right are the partial phase-shifting stripes of a white vase (top) and a white gourd (bottom) reconstructed with TVAL3 using a 15% sampling rate and four traditional Gray codes plus one complementary Gray code. It can be observed that high-quality fringe images can be obtained using TVAL3 with a low sampling rate. In the middle is the absolute phase diagram of a white vase (top) and a white gourd (bottom) reconstructed using a combination of 12-step phase-shifting stripes and four traditional Gray codes plus one complementary Gray code. It can be observed that the absolute phase of the white vase (top) and the white gourd (bottom) are correctly restored. On the far right is the three-dimensional topography of the vase (top) and gourd (bottom) reconstructed by the absolute phase combination calibration algorithm. The depth of the color represents the height information of the object, and the darker the color, the higher the height. It can be seen that our method has a good effect on the three-dimensional topography recovery of simple objects. Fig. 5(b) shows the phase unfolding process of an important line of data in a vase. The truncated phase ϕ can be expanded to absolute phase Φ by calculating k_{G1} from four traditional Gray codes and then k_{G2} from four traditional Gray codes plus one complementary Gray code. It can be seen that our experiment has correctly solved the absolute phase information.

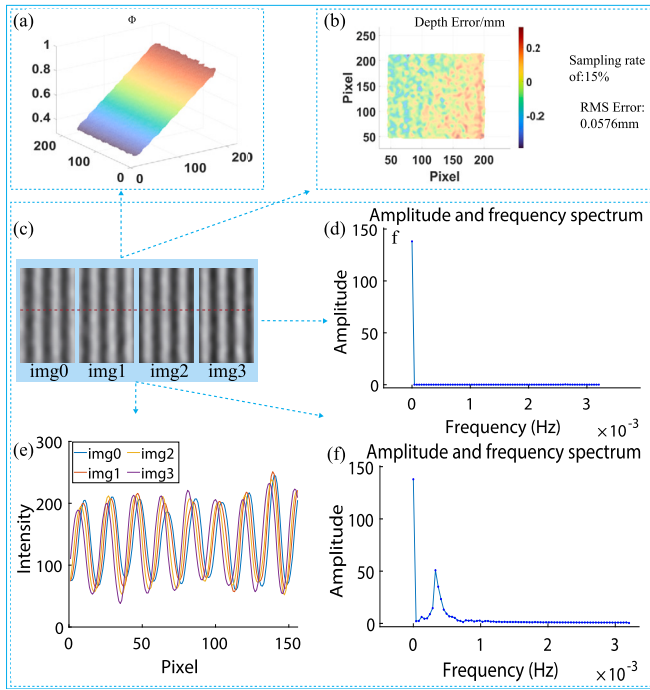


Fig. 6. Accuracy analysis of the proposed method. (a) The absolute phase map of the white plate. (b) The depth error map of the white plate. (c) The four-step phase-shift images of SPI. (d) The Fourier spectrum of the sine curve fitted to the sinusoidal graph of img0 in Fig. (c). (e) The sinusoidal graph of the cross-section in Fig. (c). (f) The Fourier spectrum plot of the sinusoidal graph of img0 in Fig. (e).

Quality evaluation of 3D SPI. To quantitatively evaluate the accuracy of the measuring system, a white plate were measured, as shown in Fig. 6. The four-step phase-shift image for SPI imaging is shown in Fig. 6(c), while the absolute phase diagram of the white plate is depicted in Fig. 6(a). It is evident that the absolute phase information of the white plate has been correctly recovered. The error diagram of the three-dimensional morphology of the reconstructed white plate is depicted in Fig. 6(b). When the sampling rate is 15%, the plane fitted by polynomials is used as the ground truth. Compared with the reconstructed white plate, the root-mean-square (RMS) errors are 0.0576 mm, with a small error, Fig. 6(d) shows the Fourier spectrum diagram of the fitted sinusoidal function of a row of data represented by the red line of img0 in Fig. 6(c). It can be seen that the fitted sinusoidal function has only one frequency, which meets the requirements of the experiment. In order to prove the sinusoidal property of fringes, the four-step phase shift is taken as an example and plot sinusoidal graphs drawn by the red lines of the four phase shift graphs in Fig. 6(c) is drawn in Fig. 6(e). It can be seen that the sinusoidal character of the fringes is moderate. Then, the sinusoidal graph in Fig. 6(e) is further transformed, and the Fourier spectrum diagram is drawn, as depicted in Fig. 6(f). According to the results, these sinusoidal fringes not only contain single-frequency components, but also multi-frequency components, possibly because the coupling between the DMD chip and the imaging lens is not perfect. The RMSE of our algorithm is only 0.0576 mm. This proves that our 3D imaging system can achieve sub-millimeter 3D reconstruction accuracy, which is acceptable for many 3D applications.

Measurement on Complex Scenes. In this experiment, to further demonstrate the effectiveness of our system, three relatively complex structures were tested: the eye, mouth, and ears of the human face, as shown in Figs. 7(a), (d), (h), respectively. The three stripe images obtained through this experimental system correspond to Figs. 7(b), (e), (i). And, the final 3D imaging results obtained are shown in Figs. 7(c), (f), (j). The results show that the detailed contour information of each

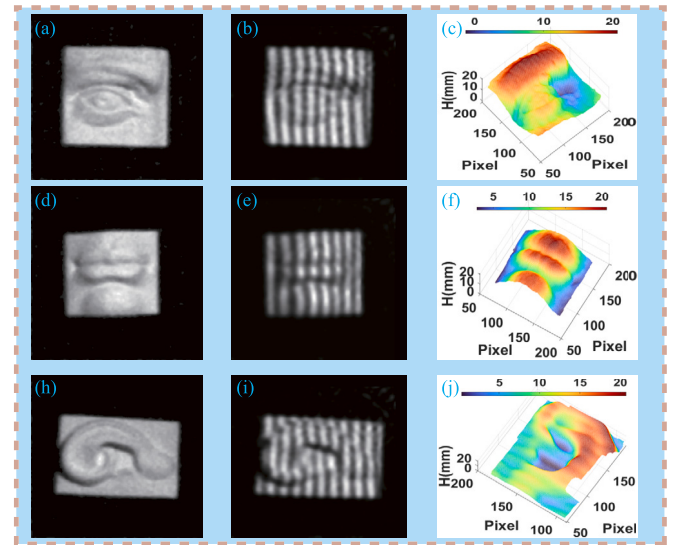


Fig. 7. The result of 3D reconstruction of facial features. (a) A two-dimensional picture of the eye. (b) The fringe pattern of the eye. (c) 3D reconstruction of the eye. (d) A two-dimensional picture of the mouth. (e) The fringe pattern of the mouth. (f) 3D reconstruction of mouth. (h) A two-dimensional picture of the ear. (i) The fringe pattern of the ear. (j) 3D reconstruction of ear.

part of the eye, mouth and ear can be clearly seen, which proves that our experimental system can reconstruct 3D shapes with high sensitivity. The application of SPI technology in 3D imaging-related fields can be further advanced by this work. It is anticipated that this technology will be widely adopted for dynamic 3D shape measurement in complex scenarios, including remote sensing, dimensional measurement, shape analysis of components in industrial assemblies, and collision testing in the automotive industry, and others in the future.

4. Conclusion

This paper proposes a quantitative optical filtering and quadrant sensing-based single-pixel 3D imaging method assisted by phase-shifted fringe projection. Using a DLP digital projector, RGB color fringes from three channels are projected simultaneously onto the target object. Meanwhile, a passive single-pixel imaging system with three-channel quadrant sensing captures the signals of the deformed fringe pattern. By combining the TVAL3 compressed sensing algorithm, the sampling rate is reduced to about 15%, significantly decreasing the sampling rate while accelerating the speed of 3D imaging. Finally, the 3D information of the object is calculated using the phase information recovered from the deformed fringe patterns and a calibrated phase-height model. It has been proven that using multiple integrated optical filtering single pixel detectors or array detectors on a single optical path, combined with multiple spatial light modulation devices, can effectively improve optical modulation and detection efficiency, and improve the speed of 3D imaging. Especially for 3D imaging technology that only uses three structured light fields, this method only requires one set of experiments to experiment with 3D imaging, which can greatly improve the speed of 3D imaging. This work can further advance the application of SPI technology in fields related to 3D imaging and is expected to be utilized in dynamic 3D shape measurement of complex scenes in the future. Potential applications include remote sensing, size measurement, and shape analysis of components in industrial assemblies, as well as collision testing in industrial vehicles. However, the proposed method still has two main disadvantages:

- Poor imaging quality of colored, gray, or black objects.* For colored 3D objects, the RGB light projected onto their surfaces cannot be

accurately distinguished by the wavelengths detected by a three-channel single-pixel quadrant detector. Furthermore, most black or dark-colored objects absorb incident light, leading to lower reflected light intensity and, consequently, reduced imaging quality.

- b. A DLP structured light projector with three internal DMDs must be equipped to achieve simultaneous acquisition of three channel data. Due to the working mechanism of digital micromirror devices (DMDs), only the binary pattern can remain unchanged at any time. Therefore, a single-pixel sensor with three-channel optical filtering and quadrant sensing must be equipped with a DLP structured light projector with three DMDs inside. To achieve simultaneous collection of three-channel data, the DLP4500 used in this research only contains one DMD internally, so this experiment is only an exemplary experiment. In the future, when using a DLP structured light projector equipped with three DMDs, it will be possible to collect data from all three channels simultaneously. This limitation should be recognized as part of this experiment.

CRedit authorship contribution statement

Baolin Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Data curation. **Xuejing Shi:** Data curation. **Cheng Zhou:** Writing – review & editing, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Binyu Li:** Software, Methodology, Data curation. **Xuan Liu:** Software, Methodology, Data curation. **Xinwei Li:** Software, Methodology, Data curation. **Jipeng Huang:** Writing – review & editing, Methodology, Investigation, Funding acquisition, Formal analysis. **Lijun Song:** Writing – review & editing, Methodology, Investigation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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